

Unit

2

Lesson

4



Innovators in Action.

Code the Conversation.

Time Frame: [80-90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Describe how energy moves through a circuit, from the power source to the glove components..
- Trace the energy transfer process that happens when a finger touches a copper strip, and a message appears on the LCD.
- Identify the independent and dependent variables in a smart glove circuit.
- Recognize different energy types.

Challenge questions of the lesson

- How does energy flow from the battery to make the glove's components work? Can you describe the path it takes through the circuit?
- What happens to the energy when a finger touches a copper strip that causes a message to appear on the LCD screen?
- In the smart glove circuit, which part is the independent variable (the one you change), and which is the dependent variable (the one that changes as a result)?
- What different types of energy can you identify in the operation of the smart glove? How do they transform from one form to another?

SEL Relation

- **Self-Awareness:** Students reflect on their own learning process by calculating energy values and understanding the impact of their actions (hand movements) in the project.
- **Self-Management:** The lesson encourages focus and persistence as students work through energy calculations and problem-solving steps.
- **Social Awareness:** Through collaboration and discussion, students develop empathy by connecting energy concepts to real human movements and assistive technology.
- **Responsible Decision-Making:** Students apply critical thinking to ensure calculations are accurate and reflect on how understanding energy helps design effective assistive devices.

Engage



Teacher's Role:

- Read the story aloud clearly, using varied tone to emphasize key moments of curiosity and discovery.
- Use gestures or visuals to help students picture the energy flow and components in the glove circuit.
- Prompt students to consider the question of how energy travels through the circuit.
- Facilitate the “Let’s Think” drawing activity by encouraging students to visualize and map the energy path.
- Support students who may need help with technical vocabulary or the concept of energy transfer.
- Encourage participation and allow students to share their drawings and ideas.
- Connect the story and drawing to prior knowledge of circuits and energy concepts.

Expected from students:

- Listen attentively to the story and engage actively with the characters’ questions and ideas.
- Reflect on the “Let’s Think” prompt and attempt to draw the energy flow from battery to LCD.
- Use labels and diagrams to show understanding of what changes and what stays constant in the circuit.
- Share ideas and drawings respectfully with peers and the teacher.
- Ask questions to clarify any confusion about the energy path.

Parts of the book included.



Let's Think!



Where does the energy start in your glove circuit? What changes along the way—and what parts depend on each other?

Draw the energy path from the battery to the LCD. Label what stays the same, what changes, and what depends on the finger touch.

Story



Objective:

1. Teacher says (with curiosity and warmth):

- “Today, Adam and Laila are reviewing their glove circuit, and they have some great questions about how energy moves inside the glove. Let’s listen and find out!”
 - (Read the story aloud with clear, expressive tone, pausing naturally at questions and key points.)

2. Teacher prompts while reading:

- When Mrs. Sara asks, “What’s the first thing the battery powers?” — pause and ask the class:
 - “What do you think? Where does the energy start its journey?”
- After Laila says, “It sends power through the copper tape...” — pause and say:
 - “Right! Can someone explain why copper tape is important here?”
- When Adam says, “Then the Arduino reads it and sends the signal to the LCD!” — pause and ask:
 - “How do you think the Arduino knows what to do with the signal?”

3. After reading:

- “Now, I want you to imagine tracing that energy path yourself. Where does it start? What parts stay the same, and what changes? What depends on what? We’ll explore that more with a drawing activity next!”

Possible Strategies for Implementation:

1. Interactive Read-Aloud:

- Use tone variation and gestures to emphasize technical terms like “energy travel,” “battery,” and “signal.”
- Invite students to act out parts of the story or use hand motions to represent energy flow.

2. Guided Questioning:

- Pause after key dialogue to ask comprehension questions such as:
 - “What do you think ‘energy travel’ means here?”
 - “Why does the battery send power through the copper tape first?”
 - “How does the Arduino ‘read’ the signal?”

3. Visual Mapping:

- Have students draw or label the energy path as they listen to reinforce understanding.
- Use whiteboards or paper for quick sketches.

4. Think-Pair-Share:

- After reading, pair students to discuss the energy flow and what depends on what, then share insights with the class.

5. Vocabulary Support:

- Pre-teach or reinforce key terms and use visual aids to support students who need it.

Let's Think!

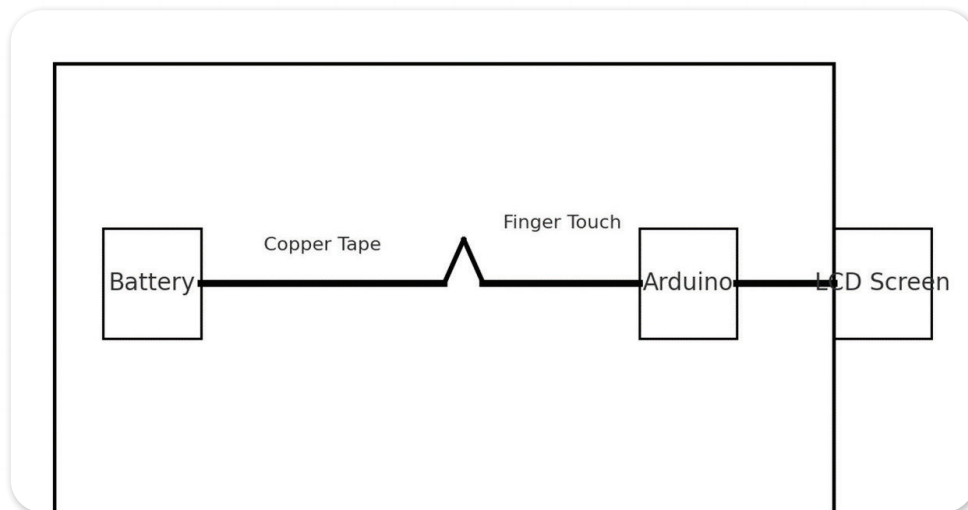


Objective:

Students will analyze and illustrate the flow of energy in the smart glove circuit from the battery to the LCD, identifying which components and energy states stay constant, change, or depend on finger touches, strengthening their understanding of energy transfer and system interactions.

Answer:

"The energy starts at the battery, which provides power continuously. It flows through the copper tape on the glove fingers to the Arduino. When a finger touches the thumb, the circuit closes, sending a signal to the Arduino. The Arduino reads the signal and sends a command to the LCD screen to display a message. The battery and copper tape's energy stay constant, but the signal changes depending on which finger touches the thumb."



Possible Strategies for Implementation:

1. Provide Clear Instructions:

- Explain the drawing task clearly, encouraging students to show energy starting at the battery, flowing through copper tape, reaching the Arduino, and resulting in an LCD message.

2. Use Visual Examples:

- Show a simple sample drawing or diagram to model expectations, highlighting arrows for energy flow and labels for components.

3. Encourage Labeling:

- Prompt students to label parts of their drawings indicating what stays the same (e.g., power source), what changes (e.g., signals sent), and dependencies (e.g., message depends on finger touch).

4. Pair or Small Group Work:

- Have students share and compare their drawings with peers to refine their understanding and discuss differences.

5. Offer Support:

- Circulate to assist students struggling with concepts or drawing skills, asking guiding questions to scaffold thinking.

6. Prompt Reflection:

- After drawing, ask students to explain their diagrams verbally or in writing to reinforce learning.

Explorer's Toolkit



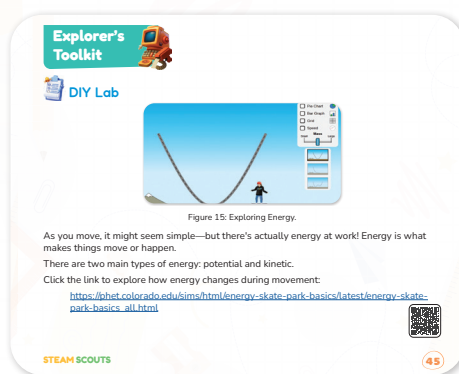
Teacher's Role:

- Introduce the concept of energy in motion, emphasizing potential and kinetic energy.
- Provide a brief overview of the PhET simulation link and explain how to navigate it.
- Guide students through the simulation's key features, such as moving the skater up and down the ramp.
- Prompt students to observe changes in energy types as the skater moves.
- Facilitate discussion around the reflection questions, encouraging students to relate the skater's energy changes to the smart glove finger movements.
- Support students needing help with the simulation interface or scientific vocabulary.
- Monitor student engagement and understanding, answering questions as they explore.

Expected from students:

- Actively engage with the PhET Energy Skate Park simulation by manipulating the skater's movement.
- Observe and note how potential and kinetic energy change as the skater moves along the ramp.
- Reflect thoughtfully on the guided questions provided in the activity.
- Make connections between the simulation and the energy transfer in the smart glove circuit.
- Participate in group discussions or share observations with the class.

Parts of the book included.



While playing, observe:

- What happens to energy when the skater is at the top of the ramp?
- What changes when the skater moves downward?
- When is the energy highest or lowest?
- How is this similar to what happens when your finger presses the glove and a message appears?

Let's Think!



Objective:

Students will investigate and describe the transformation between potential and kinetic energy during motion, and apply this understanding to explain energy changes within the smart glove circuit during finger movements.

Link: https://phet.colorado.edu/sims/html/energy-skate-park-basics/latest/energy-skate-park-basics_all.html



Answers:

- What happens to energy when the skater is at the top of the ramp?**
 - The energy is mostly potential because the skater is at a higher position with stored energy due to gravity.
- What changes when the skater moves downward?**
 - Potential energy decreases while kinetic energy increases as the skater speeds up moving down.
- When is the energy highest or lowest?**
 - Total energy remains constant (if ignoring friction), but potential energy is highest at the top, kinetic energy highest at the bottom.
- How is this similar to what happens when your finger presses the glove and a message appears?**
 - Like the skater's energy changing, pressing the glove causes energy to transfer through the circuit, triggering a message to appear.

Possible Strategies for Implementation:

1. Pre-Activity Briefing:

- Review key energy concepts (potential and kinetic) with examples before starting the simulation.

- Explain the goals of the exploration and what observations to focus on.

2. Guided Simulation Walkthrough:

- Demonstrate basic controls of the PhET simulation to ensure all students can navigate confidently.
- Model how to move the skater up the ramp and observe energy graphs or meters.

3. Think Aloud:

- Encourage students to verbalize their observations and predictions as they use the simulation.
- Pause frequently to ask guiding questions from the reflection prompts.

4. Note-Taking Support:

- Provide graphic organizers or worksheets to help students record observations systematically.

5. Peer Discussion:

- Pair students or form small groups to compare findings and discuss how energy shifts between potential and kinetic forms.

6. Make Real-World Connections:

- Facilitate conversation on how this relates to finger pressure on the glove creating signals and energy changes.



Teacher's Role:

- Introduce the video by briefly reviewing what students have learned about energy so far.
- Set a clear purpose for watching: understanding different types of energy and how they relate to the smart glove project.
- Play the video (<https://youtu.be/jhKejoBqiYc?si=mzBdP61CBl80JTvu>).
- Pause at key points to highlight important concepts or check for understanding.
- Facilitate brief discussions or questions during and after the video to reinforce learning.
- Clarify any confusing terminology or concepts for students.
- Encourage students to take notes or jot down examples of energy types mentioned.

Expected from students:

- Watch the video attentively with focus on identifying different forms of energy.
- Take notes or record examples of energy types as they watch.
- Participate in discussions or answer questions about the video content.

- Relate the information to their own work on the smart glove circuit.
- Ask questions if concepts are unclear or need further explanation.

Parts of the book included.



Screen Lessons

Tap the link to explore and learn about the different forms of energy.

<https://youtu.be/jhKejoBqiYc?si=mzBdP61CBl80JTvu>



Screen Lessons

Objective:

Students will describe and differentiate various forms of energy, including potential, kinetic, electrical, and mechanical energy, to deepen their understanding of energy concepts relevant to the smart glove's function.

Video Link: <https://youtu.be/jhKejoBqiYc?si=mzBdP61CBl80JTvu>



Possible Strategies for Implementation:

1. Pre-Viewing Activation:

- Ask students what forms of energy they already know and how they think energy might be involved in their smart glove project.

2. Segmented Viewing:

- Play the video in short segments, pausing to discuss or summarize key points before continuing.

3. Guided Note-Taking:

- Provide a graphic organizer or note prompts for students to fill in examples of different energy types.

4. Think-Pair-Share:

- After the video, have students share with a partner what new energy types they learned and how these might relate to the glove.

5. Clarification and Reinforcement:

- Review difficult concepts with simple examples or hands-on demos, if possible.

6. Connect to Project:

- Discuss how electrical energy powers the glove and how mechanical energy comes from finger movements.

Elaborate and Evaluate



Teacher's Role:

- Facilitate the Apprentice multiple-choice quiz by clarifying questions, encouraging thoughtful answers, and providing support where needed.
- Guide students through the Builder activity, helping them design and test different finger tap combinations on the glove. Prompt students to identify independent and dependent variables and the importance of controlled testing.
- Support students in the Masterminded energy calculation problems by reviewing formulas, demonstrating calculations, and assisting with problem-solving strategies.
- Encourage students to discuss their findings and reflect on how kinetic and potential energy relate to the smart glove project.
- Monitor group and individual work to provide timely feedback and scaffold learning as needed.
- Promote a growth mindset by praising effort and encouraging iterative testing and learning from mistakes.

Expected from students:

- Complete the Apprentice quiz independently, applying knowledge from previous lessons to answer questions about energy types and transformations in the glove.
- Actively participate in the Builder investigation by conducting tests, recording results, and identifying variables clearly.
- Collaborate with peers when discussing test results and share insights about experimental design.
- Work through the Masterminded math problems carefully, applying formulas to calculate kinetic and potential energy relevant to glove movements.
- Reflect on how energy concepts apply to real-world engineering problems like the smart glove.
- Take initiative to ask questions and seek help when needed.

Parts of the book included.

Assessment

Apprentice

1. Which of the following is an example of kinetic energy in the smart glove?

- The glove sitting on the table.
- The battery storing energy.
- A finger moving to tap another finger.
- The copper tape connected to the glove.

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2. What type of energy is stored in the glove's battery before it is used?

- Sound energy.
- Kinetic energy.
- Electrical energy.
- Potential energy.

3. When the glove is powered and a finger touches the copper tape, what type of energy is being transferred?

- Magnetic energy.
- Electrical energy.
- Thermal energy.
- Light energy.

4. Which statement best explains the energy transformation in the glove project?

- Movement becomes heat, then sound.
- Electrical energy changes into kinetic energy only.
- Potential energy turns into sound energy.
- Stored energy in the battery becomes electrical energy that powers the circuit.

5. Why is it important to understand kinetic and potential energy in engineering a smart glove?

- To decorate the glove with colors.
- To choose what hand to wear the glove on.
- To know how the glove stores and uses energy to work.
- To know how to sew the glove.

Builder

Can finger taps change what the glove says? Let's investigate!

Use your smart glove to test how different finger taps affect what message appears on the LCD screen.

Figure out what changes (dependent variable) and what stays the same or is controlled (independent variable).

Record Your Results:

Finger Tap Combination.	Message Shown on LCD.
Index finger only	
Thumb + Index	
Middle + Ring	
Pinky only	

Think About It:

1. What did you change on purpose in this test?
→ That's your independent variable.

2. What changed as a result of your tap?
→ That's your dependent variable.

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Mastermind

Let's Calculate Energy!

Use what you learned about kinetic and potential energy to solve the problems below:

1. A hand is moving to sign a word. It weighs 0.3 kg and moves at a speed of 2 meters per second.

How much kinetic energy does the hand have?

Your Answer:

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2. A hand is held still, 0.2 meters above the glove. Its mass is 0.4 kg.

What is its potential energy? (Use gravity = 10 m/s²).

Your Answer:

Note: Use These Formulas.

- Kinetic Energy (KE) = $\frac{1}{2} \times \text{mass} \times \text{speed}^2$.
- Potential Energy (PE) = $\text{mass} \times \text{gravity} \times \text{height}$.



Apprentice

Objective:

Students will demonstrate their understanding of kinetic and potential energy concepts and their application within the smart glove system by answering multiple-choice questions about energy types, transformations, and engineering relevance.

Answers:

1. Which of the following is an example of kinetic energy in the smart glove?

- Answer: C. A finger moving to tap another finger

2. What type of energy is stored in the glove's battery before it is used?

- Answer: D. Potential energy.

3. When the glove is powered and a finger touches the copper tape, what type of energy is being transferred?

- Answer: B. Electrical energy.

4. Which statement best explains the energy transformation in the glove project?

- Answer: D. Stored energy in the battery becomes electrical energy that powers the circuit.

5. Why is it important to understand kinetic and potential energy in engineering a smart glove?

- Answer: C. To know how the glove stores and uses energy to work.

Possible Strategies for Implementation:

1. Review Concepts Before Assessment:

- Briefly revisit key terms such as kinetic energy, potential energy, electrical energy, and energy transfer.
- Use real-life examples from the glove project to reinforce understanding.

2. Read Questions Carefully:

- Encourage students to read each question and all answer options fully before choosing.

3. Think Aloud Practice:

- Model thinking aloud for one or two questions to show how to eliminate incorrect choices.

4. Group Discussion:

- Allow students to discuss questions in pairs or small groups before final individual answers, promoting collaborative learning.

5. Provide Immediate Feedback:

- Review answers as a class after the quiz, explaining why correct answers are right and clarifying misconceptions.

6. Address Misunderstandings:

- Identify questions frequently missed and provide targeted mini-lessons or activities to strengthen those areas.



Builder

Objective:

Students will investigate how different finger tap combinations on their smart glove affect the message displayed on the LCD screen. They will identify independent and dependent variables and understand how controlled experiments work in a technology-based system.

Answers:

1. What did you change on purpose in this test?

- The finger tap combination (e.g., index only, thumb + index, etc.).
 - This is the independent variable.

2. What changed as a result of your tap?

- The message shown on the LCD screen.
 - This is the dependent variable.

3. Why do scientists test one thing at a time?

- So they can clearly see what caused the result. If they changed multiple things, they wouldn't know what made the difference.
 - This teaches controlled testing.

Activity Instructions:

1. Connect the Glove to the Arduino + LCD.

- Make sure all wires, copper tape, and pins are properly secured.

2. Turn on the system.

- Upload the Arduino sketch with the finger tap messages.

3. Introduce the Table.

- Explain how to fill in the results. Start with a demonstration: tap the index finger and show what the LCD says.

4. Test Finger Taps:

- Index finger only
- Thumb + Index
- Middle + Ring
- Pinky only
- For each combination, record the message shown.

5. Think & Reflect:

- Ask students to analyze what changed and what stayed the same.

6. Class Discussion:

- Walk through student answers to questions together. Use this as a chance to reinforce scientific language.

Possible Strategies for Implementation:

- **Activate Prior Knowledge:** Begin by reviewing the concepts of independent and dependent variables. Use relatable analogies (e.g., "What happens when you change the amount of sunlight a plant gets?").
- **Model the Setup:** Demonstrate how to safely connect the smart glove to the Arduino and LCD. Show a sample tap and what message appears.
- **Use Guided Inquiry:** Encourage students to predict what they think will happen with each combination before they test it.
- **Think Aloud:** Model thinking like a scientist: "I'll only change the finger I tap, and keep everything else the same. That way, I can see if the finger tap is what causes the message to change."
- **Pair Work:** Students work in pairs — one taps, one records. Rotate roles after 2 trials to keep engagement high.
- **Scaffold Scientific Reasoning:** Prompt with questions during testing like:

- “What’s the only thing you’re changing?”
- “What stays the same each time?”
- **Use Think-Pair-Share for Reflection:** Before discussing answers, let students reflect in writing, discuss with a partner, and then share with the class.



Mastermind

Objective:

Students will apply their knowledge of kinetic and potential energy to solve real-world problems involving hand movements in the smart glove project. They will use formulas to calculate energy values based on given mass, speed, height, and gravity.

Formulas Used:

- Kinetic Energy (KE) = $\frac{1}{2} \times \text{mass} \times \text{speed}^2$.
- Potential Energy (PE) = $\text{mass} \times \text{gravity} \times \text{height}$.
- Gravity is given as 10 m/s^2 for calculations.

Answers:

1. Problem 1:

A hand is moving to sign a word.

- Mass (m) = 0.3 kg
- Speed (v) = 2 m/s

Calculate KE:

$$\text{KE} = \frac{1}{2} \times m \times v^2 = 0.5 \times 0.3 \times (2)^2 = 0.5 \times 0.3 \times 4 = 0.6 \text{ Joules.}$$

Sample Answer:

- The hand has 0.6 Joules of kinetic energy when moving at 2 meters per second.

2. Problem 2:

A hand is positioned, 0.2 meters above the surface.

- Mass (m) = 0.4 kg
- Gravity (g) = 10 m/s^2
- Height (h) = 0.2 m

Calculate PE:

$$\text{PE} = m \times g \times h = 0.4 \times 10 \times 0.2 = 0.8 \text{ Joules.}$$

Sample Answer:

- The hand has 0.8 Joules of potential energy when held 0.2 meters above the glove.

Possible Strategies for Implementation:

1. Introduce or Review Energy Concepts:

- Quickly review kinetic and potential energy definitions and units.

- Explain the significance of mass, speed, height, and gravity in the formulas.

2. Work Through Sample Problem Together:

- Model solving one problem step by step with the class, demonstrating how to plug values into the formula.

3. Support Individual Calculations:

- Encourage students to use calculators and show their work clearly.
- Provide assistance as needed with unit conversions or arithmetic.

4. Encourage Estimation and Reasonableness Checks:

- Ask students to estimate answers roughly before calculating to check if their answers are reasonable.

5. Discuss Real-World Connections:

- Relate the calculations back to how energy affects hand movement and the function of the smart glove.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist.

- Read each statement aloud one at a time.
- After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
- Example prompts:
 - "Did you learn how people share ideas without talking?"
 - "Do you know how tools can help people talk?"

3. Student Completion

- Let students check off or color the "Now I Can" boxes they feel good about.
- Remind them: "It's okay if you didn't check all of them. We're all still learning."

4. Optional Extension

- Ask students to circle the one they're most proud of.

- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **MS-PS3-2:** Develop and use models to describe that energy is transferred from place to place by sound, light, heat, and electric currents.
- **MS-PS3-3:** Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer.

2. Common Core State Standards (CCSS) for Mathematics:

- **6.EE.B.6:** Use variables to represent numbers and write expressions when solving real-world or mathematical problems.
- **6.SP.B.5:** Summarize numerical data sets in relation to their context.

3. International Society for Technology in Education (ISTE) Standards:

- **Empowered Learner:** Students leverage technology to take an active role in choosing, achieving, and demonstrating competency in their learning goals.